

Geometry

ANSWER KEY



Area and perimeter

- 1 A rectangle has the dimensions shown below. Find its area and perimeter.

$$\text{Area} = 12 \times 7 = 84 \text{ square units. Perimeter} = 2(12 + 7) = 38 \text{ units.}$$

- 2 Find the area of a trapezoid with parallel sides 8 and 14, and height 6.

$$\text{Area} = \frac{1}{2}(8 + 14)(6) = \frac{1}{2}(22)(6) = 66.$$

- 3 A triangle has base 15 and height 10. Find its area.

$$\text{Area} = \frac{1}{2}(15)(10) = 75.$$

The Pythagorean theorem

- 4 A right triangle has the two legs shown below. Find the length of the hypotenuse c .

$$c = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15.$$

- 5 A 25-foot ladder leans against a wall, with its base 7 feet from the wall. How high up the wall does the ladder reach?

$$h = \sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24 \text{ feet.}$$

Volume and surface area

- 6 Find the volume of a cylinder with radius 4 and height 9, in terms of π .

$$V = \pi r^2 h = \pi(16)(9) = 144\pi.$$

- 7 Find the surface area of a rectangular prism with dimensions 5 by 4 by 3.

$$SA = 2(lw + lh + wh) = 2(20 + 15 + 12) = 2(47) = 94 \text{ square units.}$$

Similar triangles and proportional reasoning

- 8 Triangle ABC is similar to triangle DEF , with $AB = 8$, $DE = 12$, and $BC = 10$. Find EF .

$$\text{Scale factor} = \frac{DE}{AB} = \frac{12}{8} = 1.5. EF = 10 \times 1.5 = 15.$$

- 9 A flagpole casts a shadow 18 feet long, while a 5-foot student standing nearby casts a shadow 3 feet long. Find the height of the flagpole.

$$\frac{\text{height}}{18} = \frac{5}{3}, \text{ so height} = \frac{18 \times 5}{3} = 30 \text{ feet.}$$

Angle relationships

- 10** Two parallel lines are cut by a transversal. If one angle measures 72° , find its corresponding angle and its co-interior (same-side interior) angle.
The corresponding angle is equal: 72° . The co-interior angle is supplementary: $180^\circ - 72^\circ = 108^\circ$.
- 11** The three angles of a triangle are in the ratio 2 : 3 : 4. Find each angle.
The parts sum to $2 + 3 + 4 = 9$, and the total is 180° , so each part is 20° . The angles are 40° , 60° , and 80° .
- 12** Two angles are complementary, and one is 34° more than the other. Find both angles.
 $x + (x + 34) = 90$, so $2x = 56$, giving $x = 28$. The angles are 28° and 62° .
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Coordinate geometry: distance and midpoint

- 13** Find the distance between the points $(3, -2)$ and $(9, 6)$.
 $d = \sqrt{(9 - 3)^2 + (6 - (-2))^2} = \sqrt{36 + 64} = \sqrt{100} = 10$.
- 14** Find the midpoint of the segment connecting $(-5, 3)$ and $(7, -9)$.
Midpoint = $\left(\frac{-5 + 7}{2}, \frac{3 + (-9)}{2}\right) = (1, -3)$.
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Circles: area, circumference, and equation

- 15** A circle has the radius shown below. Find its area and circumference, in terms of π .
Area = $\pi(8)^2 = 64\pi$. Circumference = $2\pi(8) = 16\pi$.
- 16** Write the equation of a circle with center $(3, -4)$ and radius 6.
 $(x - 3)^2 + (y + 4)^2 = 36$.
- 17** A circle has the radius and shaded sector shown below. Find the area of the shaded 60° sector, in terms of π .
Sector area = $\frac{60}{360} \times \pi(9)^2 = \frac{1}{6} \times 81\pi = 13.5\pi$.
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Composite and shaded regions

- 18** A square with side length 12 has a circle of radius 5 removed from its center. Find the remaining area, in terms of π .
Square area = $12^2 = 144$. Circle area = $\pi(5)^2 = 25\pi$. Remaining area = $144 - 25\pi$.

- 19** A rectangular garden 20 m by 15 m has a circular fountain of radius 3 m built inside it. Using $\pi \approx 3.14$, find the area of the garden not covered by the fountain, to the nearest whole number.
- Garden area = 300. Fountain area $\approx \pi(9) \approx 28.26$. Remaining area $\approx 300 - 28.26 \approx 272$ square meters.
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A well-designed word problem: designing a park

- 20** This question has three parts. A city is designing a park shaped like a rectangle with a semicircular section attached to one end. The rectangular section is 30 meters long and 20 meters wide, and the semicircle is attached to one of the 20-meter-wide ends. (a) Find the radius of the semicircle. (b) Find the total area enclosed by the park (rectangle plus semicircle), in terms of π . (c) The city fences the perimeter of the park: the three straight sides of the rectangle plus the curved semicircular edge, but not the shared 20-meter edge where the semicircle attaches. Find the total fence length needed, in terms of π .
- (a) The semicircle's diameter equals the 20-meter width, so the radius is 10 meters. (b) Rectangle area = $30 \times 20 = 600$. Semicircle area = $\frac{1}{2}\pi(10)^2 = 50\pi$. Total area = $600 + 50\pi$ square meters. (c) The fenced sides are the two 30-meter lengths, the far 20-meter end, and the semicircular arc. The arc length is half the circumference: $\frac{1}{2}(2\pi \times 10) = 10\pi$. Total fence = $30 + 30 + 20 + 10\pi = 80 + 10\pi$ meters.